Transit Mobility Maps Methodology

The transit mobility maps broadly measure how far someone can travel by transit from the center of each census block in NYC. This document provides a summary of the calculation method.

Questions and comments can be sent to Transportation@planning.nyc.gov

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Data Inputs

- **Census Block:** TIGER/Line® Shapefiles Blocks (2020) https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html
- **Jobs Data:** US Census: 2023 (latest available version) LEHD Workplace Area Characteristic data (WAC) Files. Version: LODES8. S00: Total number of jobs. JT01: Primary Jobs https://lehd.ces.census.gov/data/
- **Labor Force Data:** US Census: 2023 (latest available version) LEHD Residence Area Characteristic data (RAC) Files. Version: LODES8. S00: Total number of jobs. JT01: Primary Jobs https://lehd.ces.census.gov/data/
- **Population Data**: U.S. Census Bureau. (2021). 2020 Census: United States [P1]. U.S. Department of Commerce. Retrieved from https://www.census.gov/quickfacts/fact/table/US/POP010220
- **Street Network**: OpenStreetMap (NY, NJ, CT, and PA) https://download.geofabrik.de/north-america/us.html
- **Transit Services**:
    - General Transit Feed Specification (GTFS) schedule data published by transit agencies.
    - For a small number of agencies farther from NYC where official GTFS feeds were unavailable, Transitland was used as a supplemental source. (https://www.transit.land/)
    - The model includes walking, subway, bus, rail, and ferry. For detailed information on the transit services included in the model, please see Table 1 below.

Table 1: Transit Services Included

State	Transit Service Name
New York	Amtrak
	Bee-Line Bus System
	Capital District Transportation Authority (CDTA)
	Elmira Transit System (EMTA)
	AirTrain JFK
	MTA Bus Company
	MTA Long Island Rail Road
	MTA Metro-North Railroad
	MTA New York City Transit Bus
	MTA New York City Transit Subway (plus Staten Island Railway)
	NFTA-Metro Bus
	NFTA-Metro Rail
	Nassau Inter-County Express (NICE)
	Staten Island Ferry
	NYC Ferry
	NY Waterway
	RTS / Rochester Genesee Regional Transportation Authority
	Suffolk County Transit
	Lower Hudson Valley Transit
	Ulster County Area Transit (UCAT)
New Jersey	NJ TRANSIT Bus
	NJ TRANSIT Rail
	Port Authority Trans Hudson (PATH)
	PATCO Speedline
	Seastreak Ferries
Connecticut	River Valley Transit (formerly 9-Town Transit)
	CTtransit
	Norwalk Transit District
Pennsylvania	Centre Area Transportation Authority (CATA)
	Pittsburgh Regional Transit
	rabbittransit Capital Region
	rabbittransit York
	Southeastern Pennsylvania Transportation Authority (SEPTA) Bus

State	Transit Service Name
	SEPTA Regional Rail

Note: The transit travelshed model used to produce the mobility maps does not include temporary or informal transit (e.g., Dollar Vans) or formal transit for which usable data was unavailable (e.g., NY Waterway buses). These can be important parts of the transit network. Second, as a transit model, it excludes modes like driving or biking, which are also important for many New Yorkers.

Preparation of Data Inputs

- **Street Network**:
    - Two different networks were created for different stages of the model development. An expansive "routing network" was used for routing trips. A more limited "snapping network" was created to ensure each trip started in a reasonable place.
        - Routing network: OSM PBF files for the four states—New York, New Jersey, Connecticut, and Pennsylvania—were downloaded and merged. All features were preserved in the merged routing network.
        - Snapping network: Snapping occurs when origins, represented by block centroids, are connected to starting points on the street network. For the NYC area, the snapping network filters out sidewalks and above- and below-grade street lines to improve routing connectivity and ensure more reasonable starting points.
- **Transit Services**:
    - The model includes walking, subway, bus, rail, and ferry (26 transit agencies, 34 different GTFS sources).
    - Similar to other transit documents, the results are reflective of a typical fall business weekday. Wednesday, October 29, 2025, was used; when not available, a proximate Wednesday was used.
    - The GTFS feeds were cleaned by removing invalid trips and stop-time records, fixing stop sequences, adding feed-specific prefixes to core IDs and related references to ensure that IDs are unique across feeds.

Modelling Methodology

Overview

As described below, the model produces several outcomes, all based on a calculated travelshed, which is the geographic area reachable from a given starting point within a set travel time specified by the model using one or more of the transportation modes listed above.

Model outputs for each census block:

- **Access to Land Area:**
  - Measures the acreage of census blocks whose centroids are reachable within 60 minutes of an origin census block.
- **Access to Jobs:**
  - Measures the number of jobs in the census blocks whose centroids are reachable within 60 minutes using a travel-time-based gravity approach, giving greater weight to jobs reachable in shorter travel times.
- **Access to Land Area (Limited Fare):**
  - Measures the acreage of census blocks whose centroids are reachable within 60-minutes of an origin census block, with trips costing approximately $3.10 or less.
- **Access to Jobs (Limited Fare):**
  - Measures the number of jobs in the census blocks whose centroids are reachable within 60 minutes, with trips costing approximately $3.10 or less. It uses a travel-time-based gravity approach, giving greater weight to jobs reachable in shorter travel times.

Model Setup and Parameters

- **Routing tool**
  - **The model uses** R5 (Rapid Realistic Routing on Real-world and Reimagined networks) Python version to route walking and transit trips across the network.

- **Destination Area:**
  - NY region: 16 counties across a three-state area within 25 miles of each origin census block
  - County list:
    - NY: Bronx, Kings, New York, Queens, Richmond, Westchester, Nassau, Rockland, Suffolk
    - NJ: Bergen, Hudson, Essex, Union, Middlesex, Passaic
    - CT: Fairfield

- **Origin and Destination Locations**

Broadly speaking, the model measures travel time from the center of one census block to another. The method for determining the exact start and end location of each trip is outlined below.

- Block centroids were determined by calculating the arithmetic mean position of all points in the shape.
- Occasionally a census block will have an unusual shape, such as an "L" shape, which leads the method above to put the centroid outside of the census block. When that occurs, an alternative centroid identification method is used, called "Representative Point."
- Finally, the centroids for the four census blocks listed below were manually adjusted to better reflect reasonable start/end points based on the human experience of traveling to those locations.
  - LaGuardia Airport
  - JFK Airport
  - Aqueduct Racetrack
  - St. George Terminal

- **Centroid to OSM connection snapping**
  Once centroids were determined, they needed to be connected to the street network; that snapping approach is described below:
  - The snapping method used for origin block centroids was R5py's snap_to_network() function, which finds the nearest valid location in the snapping network and designates this as the connection point to the full network.
  - The "snapping network" used for this step is described in the preparation of data inputs section of this document.

- **Departure Time:**
  - The model ran a departure every minute between 7:00am and 9:00am for every census block and then took the median isochrone from that sample of 120 isochrones to represent a typical travel distance and time for that census block.

- **Trip Types:**
  - The model includes walking-only trips and trips that combine walking and public transit.

- **Journey Settings:**
  - Maximum journey time: 60 minutes
  - Walking speed: 2.8 mile/hour (4.5 km/h)
  - Maximum walking time: 60 minutes
  - Travel time includes walking time, waiting time for transit, and time on transit. The walking time also includes estimated walking between the centroid location and the snapping point.

- **Transfers**
  - **Maximum number of transfers**
    - Unlimited fare map: Within the 60min period, up to three boardings were permitted (two transfers). For example, the following trip type was permitted: Walk>Bus>Bus>Subway>Walk
    - Limited fare map: Transfers that would have increased the cost of the trip to greater than approximately $3.10 were not permitted (see section below for more details)
  - **Time to transfer**
    - The walking time between transfers and the wait time for the next service to arrive were accounted for in the model.
    - For example, if a passenger disembarked from the A subway train and took two minutes to walk from the A-train platform to the L-train platform, and the passenger had to then wait four minutes for the next scheduled L train, all six minutes were included.
    - R5 estimates walking times between connecting transit services using modeled stop locations and the OSM pedestrian network, with an assumed walking speed of 4.5 km/h.

Model Outputs

Access to Land Area maps

- Goal: These maps measure the amount of land area reachable by walking and public transit within 60 minutes.
- Index calculation: The acreages of all census blocks reachable within 60 minutes of each block were summed to calculate the total accessible land area for each block.

Access to Jobs maps

- Goal: These maps measure access to jobs by walking and public transit using a travel-time-based gravity approach, giving greater weight to jobs reachable in shorter travel times.
- Travel-time-based gravity approach (decay function): The access to jobs score is calculated as the weighted sum of jobs reachable within 60 minutes. Jobs within 30 minutes receive full weight; those within 30–40 minutes receive half weight; those within 40–50 minutes receive one-quarter weight; and those within 50–60 minutes receive one-eighth weight. This decay function is intended to loosely represent people's preference for shorter commutes.

Limited Fare maps

- Goal: The goal of these maps is to show how transit mobility changes when a commuter only has a approximately $3.10 to spend per trip, the cost of an NYCT subway or bus fare. Therefore, certain modes or transfers aren't possible in this model because they

cost more than $3.10, such as an Express Bus Trip or a single trip involving an NYC Ferry and the subway.

- To approximate a limited fare environment, the following imperfect assumptions were used:
    - Trips on railroads, AirTrain, express buses, NJ TRANSIT buses, and other modes that typically cost over $3.10 per trip were prohibited.
    - If a trip was purely on NYC Ferry or purely on PATH it was allowed. Trips that require a transfer from NYC Ferry or PATH to another mode were prohibited.
    - For trips only involving subway, Staten Island Railway and Staten Island Ferry, two transfers (three boardings) were permitted.
    - For trips only involving subway, Staten Island Railway, Staten Island Ferry, NICE Bus, Beeline, and non-express MTA buses, one transfer (two boardings) were permitted.

Note: The above is an approximation of a limited fare environment, there are niche cases where a trip that costs more than $3.10 could happen under these assumptions and trips that cost less than $3.10 would not be permitted and should be. The $3.10 fare reflects the rate in effect as of the date of this document and may not reflect subsequent fare changes.

Post-Modeling Cleaning

Model outputs are not displayed for approximately 10 percent of census blocks. as they were challenging to score fairly due to factors such as having irregular shapes or very large sizes or representing parkland, islands, roads, bridges, or other infrastructure. However, while they were excluded as possible origins they were still accounted for in the model as possible destinations.

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